

Balance in cluster randomized trials

Gillian M. Raab^{*,†} and Izzy Butcher

Applied Statistics Group, Napier University, Merchiston, 10 Colinton Road, Edinburgh EH10 5DT, U.K.

SUMMARY

This paper explores the role of balancing covariates between treatment groups in the design of cluster randomized trials. General expressions are obtained for two criteria to evaluate designs for parallel group studies with two treatments. The first is the variance of the estimated treatment effect and the second is the extent to which the estimated treatment effect is changed by adjusting for covariates. It is argued that the second of these is more important for cluster randomized trials. Methods of obtaining balanced designs from covariates which are available at the start of a study are proposed. An imbalance measure is used to compare the extent to which designs balance important covariates between the arms of a trial. Several approaches to selecting a well balanced design are possible. A method that randomly selects one member from the class of designs with acceptable bias will allow randomization inference as well as model-based inference. The methods are illustrated with data from a trial of school-based sex education. Copyright © 2001 John Wiley & Sons, Ltd.

INTRODUCTION

Cluster randomized designs are increasingly being used for intervention studies where the decision to implement an intervention is at the level of a naturally occurring group of individuals. Examples of larger units might be communities, regions, GP practices, hospitals, clinics or schools. There is now a substantial literature on the design of cluster randomized studies [1–6]. Much of this literature has been concerned with the comparison between the commonly used paired design, where the larger units are formed into matched pairs and one of each pair is randomly allocated to each treatment, and alternative designs such as completely random allocation or stratified allocation. Relatively little attention has been paid in the literature as to how the original pairs or strata are formed. Extracting information from case studies, it appears that this is most frequently done on the basis of fairly limited administrative data, and in some instances it may be as much a convenience in organizing the study as a crucial aspect of the design. In some studies, mainly those where the unit of randomization has been large, fairly detailed background data or baseline measurements are available for the units of randomization. These data may be for individuals who will take part in the trial (for example, baseline scores), or they may consist of antecedent data (for example, outcome

*Correspondence to: Gillian M. Raab, Applied Statistics Group, Napier University, Merchiston, 10 Colinton Road, Edinburgh EH10 5DT, U.K.

†E-mail: g.raab@napier.ac.uk

measures from a previous cohort of individuals in the larger units). Alternatively they may be features defined for the larger units themselves (for example, geographically based data calculated from the location of the units). There has been some discussion in the literature on the use of covariate data in the analysis of cluster randomized trials to improve the precision of the treatment effect [6, 8].

However, where covariate data are available before the treatment allocation they can also be used to balance the arms of the study. This is the approach that was taken in the SHARE study of school-based sex education, the motivating example for our work. Other studies that have utilized this type of approach include a trial of health promotion in GP practices [9] and a trial of an innovative approach to adolescent sexual health in communities in West Africa [10].

In this paper we investigate the advantages of balancing covariates in the design of cluster randomized studies and suggest procedures that can be used to achieve balance. Our focus is particularly on studies where the unit of intervention is a large existing unit for which extensive background data are available. Questions that must be addressed include how to select the covariates for which balance is to be achieved, how to use the covariate values in deciding on the allocation of units to treatments, and how to make use of the covariate data in the analysis. Our approach is largely a model-based one, but we also consider how our proposals for the design could be incorporated into an analysis by a permutation test procedure. We illustrate our proposals using the baseline data from the SHARE trial of school-based sex education (details about the study can be accessed at the reports pages of www.msoc.gla.ac.uk).

WHY DO WE SEEK BALANCED DESIGNS?

Most of the theoretical literature discusses the design of studies in terms of the improvements in precision that can be achieved from matching or stratification to improve balance. A feature of cluster randomized trials is that the variances at the individual level and the cluster level need to be taken into account when considering the effect of adjusting for covariates on the precision of the treatment effect. In individually randomized trials the residual variance is bound to decrease when covariates are added to the model. In cluster randomized trials the variance that has the most pronounced influence on the precision of treatment effects is the between-cluster variance. This variance does not have to decrease as covariates are added to the model, although in practice it is more likely to decrease than to increase [reference [1], p. 45].

There are, however, features of the design of trials, in addition to the precision of the treatment effect, that can be equally important in ensuring that their results are accepted as a valid contribution to the evidence for decisions on health or social policy. One consequence of lack-of-balance with respect to covariates is that a crude estimate of the comparison of treatments, calculated from simple summary statistics of the individuals allocated to the two treatments, may differ from the estimated treatment effect adjusted for covariates. This has been shown to be important in studies where the random allocation resulted in a very poor balance on baseline characteristics. For example, in the Edinburgh randomized trial of breast screening [11], GP practices were randomly allocated to become screening or non-screening practices. The random allocation of 87 GP practices to be screening or non-screening practices resulted in the screening practices having less favourable values of measures of socio-economic

status. The most recent analysis of breast cancer mortality from this trial [12] gave a reduction in mortality for the screened group, rate ratio 0.87 (95 per cent CI 0.70–1.06). An analysis adjusted for socio-economic status, however, gave a larger estimate of the reduction in mortality due to screening with a rate ratio of 0.79 (95 per cent CI 0.60–1.02) and of 0.71 (95 per cent CI 0.53–0.95) when deaths more than three years after the end of the study were excluded. Even though the authors argue convincingly that the adjusted rate ratios are the more appropriate estimate, there may still be some perception by policy makers that the study presents mixed messages. Thus we suggest that the difference between the adjusted and unadjusted treatment effect is itself a design criterion that we should seek to minimize in the design of trials.

It may be easier to achieve a balanced design in cluster randomized trials, compared to those where randomization is at the level of the individual. In individual studies patients are generally recruited sequentially. Thus a design where the covariates are balanced can be achieved by a stratified randomization where the strata are defined by the ranges of the covariates, or by a sequential procedure such as minimization [13]. In cluster randomized trials the participating units are generally all assembled at the start of the study. This should allow more scope for well-designed and balanced studies, drawing on the extensive literature on the design of field or laboratory experiments. The analogy cannot be taken too far, however, since the units of randomization are generally unequal in size and are very far from being under experimental control for any aspect of the study except for the random assignment.

VARIANCE AND ADJUSTMENT CRITERIA

In this section we consider two optimal design criteria. The first is the usual criterion of the variance of the treatment effect, here denoted by VARTE. The second is a criterion defined as the difference between the crude and adjusted treatment effect and denoted here by CHATE. Once a study is completed, the variance, VARTE, can be estimated from the statistical model used for the analysis and the adjustment to the treatment effect, CHATE, can be calculated. At the design stage these quantities need to be estimated from the data available at that time. We first derive expressions for VARTE and CHATE in the context of linear models for individual and cluster randomized trials.

Individually randomized trials when coefficients are known

In a trial where individuals are randomized to treatment we consider the situation where an even number (n) of individuals are randomized to one of two treatments with $n/2$ assigned to each. As well as the outcome variable (y_i) we have measures of a series of covariates ($x_{1i}, x_{2i}, \dots$) for each assigned unit ($i = 1 \dots n$). We will assume that a linear model is appropriate and the analysis is then carried out by the analysis of covariance, so that the estimated treatment difference becomes

$$(\bar{y}_1 - \bar{y}_2) - \{\beta_1(\bar{x}_{11} - \bar{x}_{21}) + \beta_2(\bar{x}_{12} - \bar{x}_{22}) + \dots\} \quad (1)$$

when $\{\bar{y}_1, \bar{x}_{11}, \bar{x}_{21}\}, \{\bar{y}_2, \bar{x}_{12}, \bar{x}_{22}\}$ are the means of $y, x_1, x_2, \dots$ for treatment groups 1 and 2, respectively, and $\beta_1, \beta_2, \dots$ are the regression coefficients for the prediction of y from the

covariates x_1 and x_2 . The variance of the treatment effect, without adjustment for covariates, is simply

$$\sigma_e^2 \left(\frac{4}{n} \right) \quad (2)$$

where σ_e^2 is the within-treatment group variance of y . If the regression coefficients in (1) were known then the variance of the estimated treatment effect (2) becomes

$$\sigma_{e|x}^2 \left(\frac{4}{n} \right) \quad (3)$$

where $\sigma_{e|x}^2$ is the within-treatment group variance of the residuals of y in the regression of y on $x_1, x_2, \dots$. The variance of the treatment effect (VARTE) will be reduced in proportion to $(1 - R^2)$, where R^2 is the squared multiple correlation between y and the covariates. Notice that, when the regression coefficients are known, this advantage in precision is obtained for any allocation of the treatments provided covariate data are available and the model is correct. We now consider the extent to which the covariates change the estimated treatment effect. We will denote the change in the treatment effect when covariates are entered into the model by CHATE. For this linear model with known coefficients, CHATE is

$$\{\beta_1(\bar{x}_{11} - \bar{x}_{21}) + \beta_2(\bar{x}_{12} - \bar{x}_{22}) + \dots\} \quad (4)$$

which will have large values if one treatment group were predicted to have very different outcomes from the other, in the absence of a treatment effect. This criterion is important in itself since it will determine the extent to which statistical manipulation has been seen to be producing the results. Additionally, the results of trials where this effect is small are also more likely to be robust to model assumptions.

The situation where the regression coefficients are known is unlikely to occur in practice, although the standardization of patient results before analysis to adjust for features such as body size might provide one example. More commonly, the regression coefficients are estimated from the trial data, and we will see below that imbalance will affect VARTE as well as CHATE when coefficients have to be estimated.

Cluster randomized trials when coefficients are known

Before considering the estimation of regression coefficients, let us first consider how the adjustment for covariates when regression coefficients are known affects the design criteria for cluster randomized studies. Realistic examples here might be the standardization of outcomes such as event rates using national data on age-specific incidences.

For simplicity we consider the case where all clusters are the same size and there are n clusters each containing exactly m individuals. The data consist of measurements y_{ij} for the i th individual in the j th cluster. In this balanced case the adjusted and unadjusted treatment effects are exactly the same as for individual randomization, but their variances are, respectively

$$\left(\tau_e^2 + \frac{\sigma_e^2}{m} \right) \left(\frac{4}{n} \right) \quad (5)$$

and

$$\left(\tau_{e|x}^2 + \frac{\sigma_{e|x}^2}{m} \right) \left(\frac{4}{n} \right) \quad (6)$$

where τ_e^2 , σ_e^2 are the between- and within-cluster components of variance of the outcome, and $\tau_{e|x}^2$, $\sigma_{e|x}^2$ are the same components of variance for the residuals of the regression of y on the covariates. As noted by Murray (reference [1], p. 372), although the overall variance of the outcome must decrease from covariate adjustment, the expression (6) for VARTE need not decrease because the effect of covariance adjustment could be to increase the between-cluster component. In the balanced case the expression for CHATE is formally identical to the expression (4) above. However, the results are affected by the distribution of the covariates within and between the clusters. The worst case is for covariates defined at the level of the clusters, where larger values of CHATE than for individually randomized studies may be obtained. This arises because a large imbalance may be possible due to the relatively small number of clusters.

Individually randomized studies when coefficients have to be estimated

When the coefficients $\beta_1, \beta_2, \dots$ have to be estimated from the trial data, the expression (3) for the variance of the estimated treatment effect is increased by a factor that relates to the imbalance of covariates between the treatment groups. The expression for this imbalance factor is a well known result from the analysis of covariance [14, 15]. For the case of a single covariate x_1 , the expression (3) for VARTE is modified to give the following expression for the variance of the treatment effect:

$$\sigma_{e|x}^2 \left(\frac{4}{n} \right) \left\{ 1 + \frac{n(\bar{x}_{11} - \bar{x}_{21})^2}{4SS_{x1}} \right\} \quad (7)$$

where SS_{x1} is the within-treatment group sum of squares of x_1 . The expression (4) for CHATE is unchanged, except that the coefficient β_1 is replaced by its estimate $\hat{\beta}_1$. From (7) we can see that VARTE is increased by the imbalance in the covariate. In the most extreme case where there is no variation in the covariate values within treatment groups, SS_{x1} would become zero and the treatment effect would be inestimable.

Cluster randomized trials when coefficients have to be estimated

In the following we assume that the analysis of the cluster randomized trial is carried out using some form of hierarchical model. For a cluster randomized trial with n clusters of m individuals and a single covariate, the expression for VARTE is given by Raudenbush [7] as

$$\left(\tau_{e|x}^2 + \frac{\sigma_{e|x}^2}{m} \right) \left(\frac{4}{n} \right) \left[1 + \frac{n(\bar{x}_{11} - \bar{x}_{21})^2}{4 \left\{ \left(\frac{\tau_{e|x}^2}{\sigma_{e|x}^2} + \frac{1}{m} \right) SS_{wx1} + SS_{bx1} \right\}} \right] \quad (8)$$

where SS_{wx1} is the within-cluster sum of squares of x_1 and SS_{bx1} is the within-treatment group, sum of squares of the cluster means. We can see that (8) has a similar form to (7)

when there is no variation in the covariate values within clusters, remembering that n now represents the number of clusters rather than the number of individuals. This will be the case when we have cluster-level covariates such as, for example dummy variables for geographic area or measures of aggregate characteristics. If, however, covariate means are the same for both treatment groups, then there will be no increase in the variance due to the imbalance of the covariate allocation.

General expressions for VARTE and CHATE for a cluster randomized trial with unequal numbers per cluster are given in the Appendix as expressions (A3) and (A4). In these expressions differences in means become difference in weighted cluster means. If the weighted cluster means of the treatment groups are identical for all covariates, then there will be no increase in VARTE due to imbalance, and the change CHATE will also be zero. The weights in these general expressions depend on the cluster sizes and the estimates of $\hat{\sigma}_{e|x}^2$, $\hat{\tau}_{e|x}^2$.

Summary

When regression coefficients are known, VARTE is reduced by covariates that reduce the residual variance of the outcome, and for cluster randomized trials it is the between-cluster component that it is most crucial to reduce. The VARTE criterion is not affected by an imbalance in the treatment allocation. These results have been used in work that makes recommendations for sample size estimation and design of cluster randomized trials [16]. The CHATE criterion, however, is highly dependent on the imbalance of the covariates and may have greater importance for cluster randomized trials than for individually randomized trials.

When the regression coefficients have to be estimated from the data, the variance VARTE will be increased by imbalance in covariates between the treatment groups. This effect is likely to be most severe for covariates defined at the cluster level, or highly correlated within clusters. The CHATE criterion remains crucially dependent on the balance of the covariates between treatment groups. These results suggest that a general guideline for the design of cluster randomized trials is to seek treatment allocations where the differences in treatment group means of covariates to be included in the analysis are as small as possible.

All of these results are derived conditional on estimates of the variance components, and thus make no allowance for the reduction in degrees of freedom for variance estimates from the estimate of covariate regression coefficients.

CHOICE OF COVARIATES

There are two different stages in a randomized experiment when decisions must be taken as to which covariates are relevant to the study. The first stage is at the design of the study, and the second is the stage when the results are being analysed. Although the use of covariates at the design stage precedes their use at the analysis, we have seen above that the criteria for evaluating designs depend on the way that covariates are used in the analysis. Thus we will consider this aspect first.

Covariates to include in the analysis

Recently published guidelines for the analysis of clinical trials [17] as well as recent statistical work (reviewed in reference [18]), emphasize that the covariates to be included in an

individually randomized trial should be defined in the trial protocol, before the analysis is carried out, and without reference to the trial outcome data or to the way in which the randomized treatment allocation has distributed the covariates. The arguments, which hinge on the effect of model selection procedures on the validity of inferences about treatment effects, are equally applicable to cluster randomized designs.

We can see from (6) and (8) that to reduce VARTE we should select covariates that will give a small value of $(\tau_{e|x}^2 + \frac{\sigma_{e|x}^2}{m})$ compared to $(\tau_{e|x}^2 + \frac{\sigma_e^2}{m})$. Where m is large it is the between-cluster component $\tau_{e|x}^2$ that it is most important to reduce. When coefficients have to be estimated (as in (8)), a factor due to covariate imbalance can increase VARTE. Once the decision to include a covariate has been made, this imbalance factor is not dependent on how well the covariates included predict the outcome. The criterion CHATE depends crucially on both the imbalance factor and on the size of the regression coefficients for predicting the outcome from the covariates in the final model used in the analysis. Thus for both of the criteria VARTE and CHATE we would like to identify a set of covariates with a strong relationship with the outcome measure at the cluster level, to include in the analysis.

Covariates to balance a design

It is obvious from the results above that we should attempt to balance a design by having identical treatment group means for all variables to be included in the analysis. There are various reasons why this may not be possible. First, for the relatively small number of clusters that are usually available, it may not be possible to obtain an exact balance on all relevant covariates. Secondly, although the covariates in a trial need to be measured before the intervention, the covariate data may not have been collected when the decision on treatment allocations has to be made. Thus the covariates that are used to balance the design may have to be surrogates for those that will be collected at baseline and that will subsequently be used in the analysis. Additionally, there may be other features of the clusters which it is thought advisable to balance in the analysis, although they are unlikely to be used in the analysis. An example of this could be geographic or administrative groupings, where policy changes during the conduct of the trial might affect the outcome for all clusters in each grouping. An example of this from the SHARE study was the local authority where each school in the study was located. It was considered important that the allocation of schools was such that there was a balance of control and intervention schools within each local authority group.

Thus, to summarize, at the stage of designing a cluster randomized trial we should identify those covariates that we are likely to use as covariates in the analysis and also any other covariates that it is possible may have an effect on the outcome (for example, administrative regions). If baseline data are not available at the planning stage, then surrogate measures should be obtained. The allocations of clusters to treatments should then be such as to achieve a balance in the means of the covariates for the clusters assigned to each treatment.

DEFINING A BALANCE CRITERION

At the stage of designing a study we are likely to have some data available on the clusters that consist of either baseline data that will be used in the analysis of the study, or surrogate indicators for other factors that may influence the outcome of the study. Some procedure must

then be used to obtain a balanced outcome on all of these factors. If we are seeking designs that will give a low value of VARTE then we will obtain the best balance by minimizing the expression for VARTE given in the Appendix. Provided we have an estimate of the ratio $\tau_{e|x}^2/\sigma_{e|x}^2$, that we can obtain from an estimate of the interclass correlation adjusted for covariates, then we can identify the design with the best variance properties. The situation for CHATE is different, however, because we cannot identify the design with the lowest CHATE without assumptions about the relationship between the outcome and the covariates, and good estimates of the coefficients $\hat{\beta}_1, \hat{\beta}_2, \dots$ that will be obtained at the analysis of the trial are unlikely at the planning stage. In particular, estimates of the partial coefficients that will arise in a particular study would require that we had data from previous trials that had used exactly the same set of covariates in a study population similar to the current one. In order to overcome these difficulties we propose using a simple balance criterion that seeks a balanced allocation of cluster means for all covariates. Such a criterion can be defined as

$$B = \{w_1(\bar{x}_{11} - \bar{x}_{21})^2 + w_2(\bar{x}_{12} - \bar{x}_{22})^2 + \dots\} \quad (9)$$

where the weights (w) determine the relative contribution of each covariate to the balance (B). A simple choice of weights would be to take the weight for the i th covariate as the inverse of the variance of the cluster means. This would be equivalent to standardizing the cluster means so that their variances across clusters were all unity. An informed choice of weights might be used if certain covariates were thought, from other evidence, to be likely to be more strongly related to the outcome than others. An advantage of the criterion (B) is that it can equally well be calculated from surrogate measures of the covariates that will be used to adjust the analysis.

The criterion (B) does not include any measure of the number or size of units that are to be assigned to the two treatment groups. It is assumed that the number of clusters assigned to each treatment group will be approximately equal and potential designs evaluated by the criterion (B) would be restricted to those where this was the case. When the clusters vary greatly in size there is also the question of balancing the total numbers of individuals in the treatment groups. Several approaches are possible here. The size of the units could itself be considered as a covariate and incorporated into the expression (9) for B . Designs could be restricted to those where the numbers of individuals on the treatments did not differ by more than an allowed number. A more sophisticated approach might use an estimate of the intracluster correlation coefficient along with the number of members of each cluster to define cluster weights, following the development in the Appendix.

A simple approach to achieving a balance for the covariates in the treatment allocation was used in designing the SHARE study. The next section will give details and evaluate the extent to which this appears to have been successful, using some baseline data from the study as if it were outcome data.

Given a criterion to choose between designs, there still remains the problem of finding a design that will have a low value of B . If the number of clusters is small then a total enumeration of all possible designs may be feasible. Other approaches might include some forms of sequential allocation and *ad hoc* procedures. Sequential allocation has been used in a study of health promotion for GP practices [9]. An example of an *ad hoc* procedure that was used in the SHARE study was to start with a possible design and then attempt to reduce the value of B by reversing the allocation of pairs of units in different clusters.

Once two possible sets of clusters have been identified, then the decision as to which treatment is to be allocated to which set of clusters can be made by a random mechanism. This fulfills the conditional independence criterion that is required for a model-based analysis of a trial, but it does not provide the basis for any form of randomization inference. If the option of a randomization analysis is considered important, then the criterion B could be used in the following manner. A maximum allowable value of B would be defined, and only those treatment allocations that give values of B below the criterion should be considered. One allocation from this possible set of designs should then be selected at random. Randomization inference could then proceed straightforwardly by judging the actual outcome, using any analysis thought appropriate, in relation to the possible outcomes from the set of potentially balanced designs. The value of B should be chosen so that the set of allowable designs is sufficiently large for inferences to be reliable. This procedure would be in the spirit of what Tukey has recently described as ‘tightening the clinical trial’ [19].

ILLUSTRATIVE EXAMPLE

The SHARE project has involved developing and then evaluating (by a cluster randomized trial) a sex education programme for use in secondary schools. SHARE, which stands for SEXUAL HEALTH AND RELATIONSHIPS, SAFE HAPPY AND RESPONSIBLE, was first developed with funding for the Health Education Board for Scotland. The education package is based on insights from sociology and psychology and experience in other countries, especially the Netherlands, about what motivates young people. After development and pilot work in several schools the package was deemed ready for evaluation. The intervention consisted of giving the schools SHARE material and also providing resources that would cover replacement staff costs to allow staff who deliver sex education in the school to be trained in the use of the SHARE material. The schools in the control arm of the study received monetary compensation equivalent to the resources that they would have received had they been an intervention school. The intervention is now complete, baseline data have been collected and we are in the process of collecting the final set of follow-up data.

The major outcomes of the study are self-reported sexual experiences and attitudes, in the short term, and young women’s reproductive outcomes, in the longer term. Sample number calculations, making allowance for an estimate of the intercluster correlation coefficient, indicated that a study that randomized approximately 12 schools to each of the arms of the trial would be required. The study is set in an area of South-East Scotland, and data from non-denominational secondary schools in the area were collected to plan the trial. Letters were sent to all head teachers in the area and, after various negotiations, a total of 25 schools, which it was estimated would provide the required pupil numbers, agreed to take part. All schools agreed on the basis of being willing to be either a control or an intervention school. Background measures were available about these schools from various sources, see Table I.

Information on location and local authority grouped the 25 schools into five strata containing 8, 2, 2, 4 and 9 schools. It was decided to balance the allocation within these strata, by assigning the two treatments within strata in the ratios 4:4, 1:1, 1:1, 2:2 and 4:5. In addition, only four schools had more than 4 per cent of their pupils from Asian ethnic groups, and the design was to be constrained to have two of these in each arm of the trial. From the remaining measures in Table I (items 4 to 15) a balance criterion (B) was calculated by summing the

Table I. Covariate data available to balance the design of the SHARE study.

Measure	Source of data
1. Location	
2. Local authority	
3. Proportion of pupils of Asian origin	Local authorities
4. Number of pupils estimated to be in schools for SHARE cohorts	Local authorities
5. Proportion of placing requests into school	Local authorities
6. Proportion of pupils eligible for free school meals	Local authorities
7. Proportion of pupils of year 4 pupils staying on at school to year 5	Local authorities
8. Proportion of year 5 pupils staying on to year 6	Local authorities
9. Attendance	Scottish Executive
10. School leavers who are unemployed or on government training schemes	Scottish Executive
11. Unemployment in the catchment area	1991 census
12. Carstairs deprivation score for the catchment area	1991 census
13. Time for and quality of sex education in the school curriculum	Scored from data collected by researchers
14. Accessibility of a family planning clinic to young people in the schools	Information collected on clinic opening hours and travel times to clinics
15. School ethos	Scored from views of professionals familiar with all schools

squares of the differences in the means of the measures, each one being standardized to have a zero mean and unit standard deviation. In addition we imposed the constraint that the cohort numbers should not differ by more than 200 between the arms of the trial. Possible allocations of the schools into treatment groups were then investigated via a search procedure and one that provided a low value of B was chosen. The allocation of the two groups to intervention and control arms was then done by the toss of a single coin at a meeting where school representatives were invited to be present.

We will illustrate how this process has worked by using the baseline data that have already been collected as though they were outcome data. One of the schools would not agree to provide baseline data, considering the topic too sensitive for 13/14 year olds. They have remained in the trial and are providing follow-up data. For the purposes of illustrating the methodology, we have omitted this school along with another school (also in the stratum of 9) that was a good match to the school with missing data and that was allocated to the opposite arm in the actual trial. Respecting the balance by measures 1 to 3 we then have a total of 67 200 possible allocations of the 23 schools. The distribution of the balance criterion B for every possible allocation is illustrated in Figure 1.

We can see that it has a highly skewed distribution, with a large number of acceptable possible designs. The actual allocation chosen gave a value of B of 0.61 which lies at the 10th percentile of this distribution.

Baseline data have been collected on sexual attitudes and behaviour from these 23 schools. The measure that we have chosen to illustrate the methodology is the answer to questions about the extent to which the young person is confident that they will be able to use a

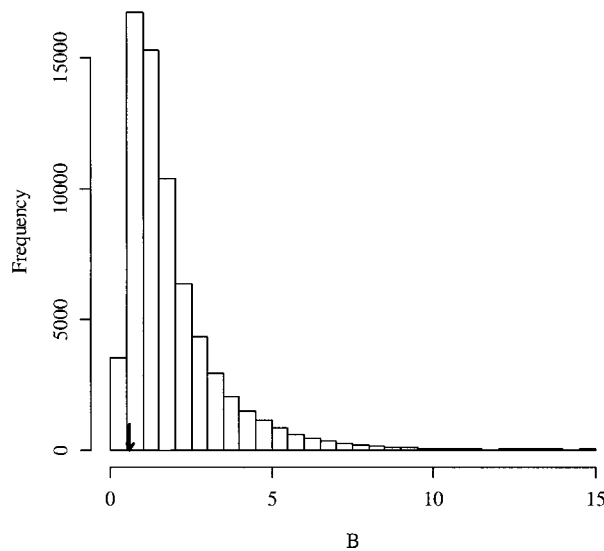


Figure 1. Distribution of imbalance criterion (B) over all 67 200 possible realizations of the trial with different treatment groups. The arrow shows the position of B for the actual allocation in the SHARE study.

condom effectively. It is denoted by CSE (condom self-efficacy) and measured as a mean of two questions each on a five point scale, where 1 = 'not all confident' to 5 = 'very confident'. This is one of several measures that we plan to use as outcome measures from the follow-up data. Because the data we are using here were collected before the intervention we know that the true effect of treatment group is zero. We also selected a set of covariates, from the other data collected at baseline, that would be a possible set for which adjustment might be expected in analysing the trial. These covariates are gender, amount of weekly pocket money, type of housing, family structure, religious belief, age of mother and a measure of parental protectiveness. The results of the prediction of the CSE score from these covariates are given in Table II.

A group of treatment allocations was selected by identifying the allocations for a set of percentiles from the distribution of B . The data from each possible realization of the trial were then analysed assuming a classical variance components model using SAS PROC MIXED [20] and the estimates of the quantities CHATE and VARTE obtained. The results are given in Table III.

We can see that the effect of adjusting for covariates on improving VARTE (shown in the table in terms of the standard error of the treatment effect) is minimal. However, the effect on CHATE can be quite considerable, with quite irregular results across the range of B , the lowest percentiles having lower values of CHATE. The balance criterion B here was calculated from the school-level covariates that were available at the time of the treatment allocations, whereas the adjustment for covariates was done for individual-level covariates collected at baseline. It seems likely that the irregular values for CHATE in Table II arise from the lack of correlation at a school level between these two sets of covariates. These

Table II. The prediction of the outcome (CSE) from individual level covariates.

Regression coefficients	Estimate	Standard error	<i>t</i>	Pr > <i>t</i>
Intercept	2.002			
Parental protectiveness (0–12)	0.042	0.0033	12.51	0.0001
Sex:				
female	−0.326	0.0178	−18.31	0.0001
male (baseline)	0.000			
Family structure:				
father only	−0.153	0.0450	−3.39	0.0007
mother only	−0.125	0.0206	−6.05	0.0001
missing	−0.455	0.1350	−3.37	0.0008
neither parent	−0.227	0.0882	−2.57	0.0101
both parents (baseline)	0.000			
Pocket money:				
under £5	0.147	0.0238	6.20	0.0001
£10 to under £15	−0.138	0.0224	−6.15	0.0001
£15 or more	−0.220	0.0245	−9.00	0.0001
£5 to under £10 (baseline)	0.000			
Housing tenure:				
care or foster home	0.094	0.1628	0.58	0.5624
council or rented	0.072	0.0207	3.48	0.0005
don't know or missing	0.145	0.0302	4.80	0.0001
privately owned (baseline)	0.000			
Mother's age:				
missing	0.045	0.0263	1.73	0.0842
under 40	−0.069	0.0193	−3.58	0.0004
over 40 (baseline)	0.000			
Religious belief:				
missing	0.115	0.0702	1.64	0.1016
not religious	0.092	0.0212	4.31	0.0001
religious/very religious	0.156	0.0313	4.98	0.0001
unsure	0.153	0.0235	6.53	0.0001
not at all (baseline)	0.000			
Variance components:				
school	0.010			
residual	0.522			

issues deserve further investigation both for this trial and in terms of developing guidelines for other studies.

We can see from these results how we might use the data to obtain a randomization distribution for analysing the results of such a trial. If the procedure of investigating the distribution of B were carried out at the design stage, then the actual design could be selected randomly from among the 10 000 (say) allocations that give the lowest value of B . The actual trial would then be analysed using any appropriate methodology. Exactly the same methodology would then be applied to the data with the treatment effect taken as one of the other 9999 allocations that were not chosen. The position of the actual estimated treatment effect in the distribution of the 10 000 that could have been chosen would then provide a p -value for a randomization test.

Table III. Effect of inclusion of covariates on the estimated treatment effect for treatment allocations at the percentiles of B .

Percentile	B	Standard error of treatment effect			CHATE
		Unadjusted	Adjusted	Difference	
Lowest	0.11	0.0493	0.0473	0.0020	0.0023
1%	0.32	0.0490	0.0466	0.0024	0.0140
2.5%	0.40	0.0492	0.0472	0.0020	0.0029
5%	0.49	0.0483	0.0446	0.0037	0.0264
7.5%	0.56	0.0489	0.0457	0.0032	0.0245
10%	0.61	0.0492	0.0468	0.0024	0.0113
25%	0.90	0.0492	0.0473	0.0019	0.0040
Median	1.42	0.0488	0.0467	0.0021	0.0046
75%	2.32	0.0491	0.0460	0.0031	0.0270
95%	4.83	0.0470	0.0467	0.0003	0.0328
99%	7.35	0.0489	0.0469	0.0020	0.0590
Highest	14.56	0.0477	0.0472	0.0005	0.0821

DISCUSSION

In this paper we suggest that in balancing covariates in the design of a cluster randomized trial, the most important factor to consider is the extent to which the adjustment will affect the estimated treatment effect. In the study that we use to illustrate our results, the effect of adjusting for covariates on the standard error of the treatment effect is minimal. This is in agreement with what has been found for other studies. A poor allocation of treatments to units could, however, produce a substantial difference between the unadjusted and adjusted estimates of the treatment effect that would have implications for the interpretation of the results.

We have presented general expressions for the adjustment to the treatment effect (CHATE) and for the variance of the treatment effect (VARTE) that apply to the unbalanced case with many covariates. This should allow data from real trials to be used to investigate the properties of different methods of balancing a design. For example, the methodology used in reference [9] where the clusters were entered into a minimization algorithm in some random order might be an alternative to the methods proposed in this paper.

The development we have used here makes no allowance for the imprecision introduced by the estimation of the variance components. This may well be important for trials where the number of clusters is small. Other authors have shown that this can be an important aspect of cluster randomized designs [6]. There are various ways that our results could be extended to incorporate this aspect of trial designs. One that we consider interesting would be to incorporate some prior information into our estimates of the variance components based on the recent empirical work on the size of intraclass correlations [21, 22].

APPENDIX

For a trial with n_1 clusters allocated to treatment 1 and n_2 clusters to treatment 2 we have an outcome y_{ij} for the i th member of the j th cluster and the indicator t takes the values 1 or

2 accordingly as the j th cluster is allocated to treatment 1 or to treatment 2. In addition we have p covariates and the value for the k th covariate for the i th member of the j th cluster is x_{kij} . The number of members of the j th cluster is m_j .

A hierarchical model is assumed with between- and within-cluster variances τ^2 and σ^2 . The estimate of the treatment effect, without any adjustment for covariates, becomes a difference in weighted cluster means

$$y_{..2} - y_{..1} = \frac{\sum_{j:t=2} w_j y_{.j}}{\sum_{j:t=2} w_j} - \frac{\sum_{j:t=1} w_j y_{.j}}{\sum_{j:t=1} w_j} \quad (\text{A1})$$

where the weights are $w_j = s^2 m_j / (s^2 + t^2 m_j)$ and s^2 and t^2 are the estimates of σ^2 and τ^2 .

The variance of the estimated treatment effect (A1) can be estimated by

$$s^2 \left(\frac{1}{\sum_{j:t=1} w_j} + \frac{1}{\sum_{j:t=2} w_j} \right) \quad (\text{A2})$$

When the analysis is adjusted for p covariates the change in treatment effect (CHATE) becomes

$$\sum_{k=1, \dots, p} \hat{\beta}_k (x_{k..2} - x_{k..1}) \quad (\text{A3})$$

where $\hat{\beta}_k$ is the estimate of regression coefficient for predicting y from the k th covariate and $x_{k..t}$ is the weighted mean of the cluster means for treatment t for the k th covariate, with weights w_j as defined above. The expression for the estimated variance of the treatment effect becomes

$$s^2 \left(\frac{1}{\sum_{j:t=1} w_j} + \frac{1}{\sum_{j:t=2} w_j} \right) \left\{ 1 - \frac{1}{4} \left(\frac{1}{\sum_{j:t=1} w_j} + \frac{1}{\sum_{j:t=2} w_j} \right) Q \right\}^{-1} \quad (\text{A4})$$

where s^2 and the w_j are estimated from the model including covariates and Q is a quadratic form $(\Delta X)'(\text{SSX})^{-1}(\Delta X)$ with ΔX consisting of a column vector of differences in weighted sums of the covariate cluster means for the two treatments with element $(\sum_{j:t=2} w_j x_{k.j} - \sum_{j:t=1} w_j x_{k.j})$, and SSX is a matrix made up of sums of squares and products of the x 's within and between clusters. The $k1, k2$ element of SSX has the form

$$\sum_i \sum_j (x_{k1ij} - x_{k1..j})(x_{k2ij} - x_{k2..j}) + \sum_j w_j (x_{k1..j} - x_{k1..})(x_{k2..j} - x_{k2..})$$

where $x_{k..j}$ is the mean of the k th covariate for the j th cluster and $x_{k..}$ is the overall weighted mean of the k th covariate.

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REFERENCES

1. Murray DM. *Design and Analysis of Group-randomised Trials*. OUP: New York, 1998.
2. Donner A. Some aspects of the design and analysis of cluster randomization trials. *Applied Statistics* 1998; **47**(1):95–113.
3. Green SB, Corle DK, Gail MH, Mark SD, Pee D, Freedman LS, Graubard BI, Lynn WR. Interplay between design and analysis for behavioural intervention trials with community as the unit of randomisation. *American Journal of Epidemiology* 1995; **142**:587–593.
4. Gail MH, Mark SD, Carroll RJ, Green S, Pee D. On design considerations and randomisation-based inference for community trials. *Statistics in Medicine* 1996; **15**:1069–1092.
5. Donner A, Klar N. Cluster randomization in epidemiology: theory and application. *Journal of Statistical Planning and Inference* 1994; **42**:37–56.
6. Diehr P, Martin DC, Koepsall T, Cheadle A. Breaking the matches in a paired t-test for community interventions when the number of pairs is small. *Statistics in Medicine* 1995; **14**(13):1491–1504.
7. Raudenbush SW. Statistical analysis and optimal design in cluster randomized trials. *Psychological Methods* 1997; **2**(2):173–185.
8. South MC, Day NE. Cluster randomisation in large public health trials: the importance of antecedent data. *Statistics in Medicine* 1992; **11**(3):307–316.
9. Hilton S, Doherty S, Kendrick T, Kerry S, Rink E, Steptoe A. Promotion of healthy behaviour among adults at increased risk of coronary heart disease in general practice: methodology and baseline data from the Change of Heart study. *Health Education Journal* 1999; **58**:3–16.
10. Todd J, Hayes R, Ross D, Balira R, Grosskurth H, Mosha F, Changalucha J. Using an initial HIV and chlamydia prevalence survey to increase the power of a community randomised trial in rural Tanzania. Abstract, Thirteenth Meeting of the International Society for Sexually Transmitted Diseases Research, Denver, USA, 11–14 July 1999.
11. Alexander F, Roberts MM, Lutz W, Hepburn W. Randomisation by cluster and the problem of social class bias. *Journal of Epidemiology Community Health* 1989; **43**:29–36.
12. Alexander FE, Anderson TJ, Brown HK, Forrest AP, Hepburn W, Kirkpatrick AE, Muir BB. 14 years of follow-up from the Edinburgh randomized trial of breast-cancer screening. *Lancet* 1999; **353**(9168):1903–1908.
13. Pocock SJ, Simon R. Sequential treatment assignment with balancing for prognostic factors in the controlled clinical trial. *Biometrics* 1975; **31**:103–115.
14. Cox DR, McCullagh P. Some aspects of covariance analysis. *Biometrics* 1982; **38**:541–561.
15. Stuart A, Ord JK. *Kendall's Advanced Theory of Statistics Volume 2*. Edward Arnold: London, 1991; section 28.46, p. 1066.
16. Murray DM, Hannan PJ. Planning for the appropriate analysis in school-based drug-use prevention studies. *Journal of Consulting and Clinical Psychology* 1990; **58**(4):458–468.
17. International Conference On Harmonisation of Technical Requirements For Registration of Pharmaceutical For Human Use — ICH Harmonised Tripartite Guideline. Statistical principles of clinical trials. *Statistics in Medicine* 1999; **18**:1905–1942.
18. Raab GM, Day S, Sales J. How to select covariates to include in the analysis of a clinical trial. *Controlled Clinical Trials* 2000; **21**:330–342.
19. Tukey JW. Tightening the clinical trial. *Controlled Clinical Trials* 1993; **14**:266–285.
20. SAS Institute. SAS Technical Report P-229, SAS/STAT Software: Changes and Enhancements, Release 6.07. SAS Institute Inc: Cary, NC, 1992.
21. Gulliford MC, Ukoumunne OC, Chinn S. Components of variance and intraclass correlations for the design of community-based intervention studies. *American Journal of Epidemiology* 1999; **149**:876–883.
22. Murray DM, Short BJ. Intraclass correlations among measures related to tobacco use by adolescents: estimates, correlates, and applications in intervention studies. *Addictive Behaviors* 1997; **22**(1):12.